Research gap and claim boundary

Research gap and claim boundary I

The two communities of sec. 2 provide substantial pieces of the problem, but the cited threads do not answer the same question. The gap is not a claim that either field is incomplete; it is the untested intersection between active-inference belief sharing and robust generalized Bayes. This section describes that intersection thread by thread, states the scoped bridge evaluated here, and records the evidence boundary that travels with it.

Five threads in the reviewed literature run toward the gap but do not, in the sources cited here, cross it — three from active inference and two from robust Bayes.

Thread 1 — Generative modeling and action (active inference). Discrete active inference is a mature, synthesized formalism with standardized tensors and an expected-free-energy action rule (Friston 2010; Da Costa et al. 2020), and it is executable at scale through pymdp (Heins et al. 2022) and RxInfer (Bagaev et al. 2023). *Boundary:* these sources do not by themselves equip the active-inference inference step with the contamination mechanism evaluated here.

Thread 2 — Collective belief coordination (active inference). Colonies coordinate by sharing beliefs and minimizing collective surprise, reproducing flocking (Heins et al. 2024), epistemic-community formation (Albarracin et al. 2022), and collective intelligence (Kaufmann et al. 2021). *Boundary:* the cited collective-belief studies treat the broadcast posteriors as trusted and do not evaluate the robustness of fusion to misspecified or intentionally wrong members.

Thread 3 — Belief sharing and structure growth (active inference). Federated belief sharing fuses posteriors into a hive-mind consensus (Friston et al. 2024), and post-hoc Bayesian model reduction (Friston and Penny 2011) together with optimal-design-flavored model selection (Smith et al. 2022) grow and prune the generative model. *Boundary:* in the cited belief-sharing line, fusion is exact-Bayes and trusting; its connection to the robustness theory developed for federated learning is not evaluated.

Thread 4 — Federated and partitioned inference (robust Bayes). Federated learning aggregates decentralized models (McMahan et al. 2017), and partitioned variational inference gives the factor-algebra framework that unifies federated and continual learning (Bui et al. 2018; Ashman et al. 2022). *Boundary:* the cited examples target parameter or predictive-model factors rather than the generative-model-bearing, action-selecting POMDP belief consensus used here.

Thread 5 — Generalized and robust Bayes (robust Bayes). Generalized Bayesian updating replaces the likelihood–KL pair with a loss–divergence pair (Bissiri et al. 2016; Knoblauch et al. 2022); bounded losses — the density-power β-divergence (Basu et al. 1998) and generalized cross-entropy (Zhang and Sabuncu 2018) — deliver bounded influence; and FedGVI (Mildner et al. 2025) federates the robust objective with provable guarantees. *Boundary:* the cited robust-Bayes apparatus does not evaluate active-inference POMDP belief consensus. Its behavior in the discrete categorical regime of the worked belief-sharing example (Friston et al. 2024) is the scoped setting evaluated here.

The belief-fusion bridge evaluated here I

Across these five reviewed threads, the missing intersection is specific: the active-inference sources provide belief fusion but not the contamination analysis used here, while the robust-Bayes sources provide robust inference but not this acting-agent categorical consensus. We evaluate the bridge with robust, generalized-Bayes belief fusion for active-inference ensembles, comprising three components and one recovery anchor. (i) A per-agent, FedGVI-faithful generalized-Bayes update carrying bounded-influence robustness through the β- and rcce-losses (eq. 4, eq. 5). (ii) A complementary server-side divergence-reweighting *heuristic* that discounts each agent by its divergence from the emerging consensus (eq. 7). (iii) A conservative server-side variational rule with a stated aggregation free energy and a redescending raw effective-weight bound (eq. 8).

The shared anchor is recovery of the standard-Bayes client limit and the project log-linear-pool server identity in their trusting limits (sec. 6.1). Under the qualified bridge of sec. 5.8, the latter specializes Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024), not the complete source protocol. The algebraic result and machine-precision checks therefore identify a recovery boundary rather than a replacement. Headline comparisons use matched paired statistics — paired Wilcoxon (Wilcoxon 1945), Benjamini–Hochberg FDR (Benjamini and Hochberg 1995), bootstrap confidence intervals (Efron and Tibshirani 1993), and observed-effect design-power planning — and are certified for reproducibility (Peng 2011).

Guarantee map: three robustness axes I

A red-team review surfaced a distinction we carry through the paper rather than paper over: robustness enters in **three** places, and they do not have the same theoretical standing.

Guarantee map: three robustness axes I

1. **Client-side (rigorous).** The per-agent generalized-Bayes update, driven by a bounded loss inside `generalized_posterior` (eq. 4, eq. 5). It is derived from the stated objective eq. 1 and provably limits to negative-log-likelihood / Bayes — and hence to the standard pool — as the loss parameter goes to zero (Corollary 6 + Proposition 4). This axis inherits FedGVI's (Mildner et al. 2025) bounded-influence result only under the source theorem's stated loss, model, and contamination assumptions.
2. **Server-side (heuristic).** The divergence-reweighting aggregator `robust_aggregate`, which discounts each agent by $\exp(-c\,\mathrm{KL}(q_n \,\|\, q))$. Only its *recovery* limit is proven — at $c = 0$ it equals the standard log-linear pool (eq. 7, Theorem 5); it is **not** the closed-form minimizer of a FedGVI objective. We present it as a complementary heuristic and never claim it inherits the bounded-influence bound.
3. **Server-side (objective-backed, conservative).** The variational aggregator `variational_aggregate` applies exact block updates that do not increase the stated free energy eq. 8, recovers the same log-linear pool in the trusting limit, and bounds each raw effective weight by its base weight. Its honest cost is conservatism: it is not the sharp accuracy-maximizing heuristic.

The robustness sweep (sec. 7.5) reports these axes, and sec. 8.15 states which claim rests on which. No figure, table, or sentence in this paper grants the server-side heuristic the guarantee that belongs to the client-side update or the variational server objective: the effect-size, confidence-interval, and power enrichment that decorate the sweep characterize the heuristic's *behavior*, not a per-agent or variational guarantee. This honesty is the point: the client-side axis is source-theorem-backed, the sharp server heuristic is useful but clearly labeled, and the variational server axis is rigorous but conservative.